Paper 1: Medicine Through Time with the Western Front

High-Yield Revision Cheat Sheet

1. Master Glossary

Term	Definition
Animalcules	The term used by Antony van Leeuwenhoek in 1683 to describe the tiny, microscopic living organisms (bacteria) he observed under his improved microscope, which marked the first recorded observation of bacteria but was not yet linked to causing disease.
Antibiotic	A drug made from microorganisms, such as mould, that can destroy or inhibit the growth of harmful bacteria inside the body to cure infections.
Antibiotics	Medicines, such as penicillin, that destroy or inhibit the growth of bacteria inside the body to cure infectious diseases, transforming healthcare after their mass production during WWII.
Aseptic surgery	A surgical method pioneered by c1900 where infection is prevented by ensuring the entire operating environment is completely free of bacteria through steam-sterilising all instruments, washing hands, and wearing sterile gowns and rubber gloves.
Bacteriology	The scientific study of bacteria, pioneered by Robert Koch who developed rigorous methods to grow, stain, and identify the specific microbes causing individual killer diseases.
Bills of Mortality	Weekly published sheets in London listing the numbers and causes of deaths in each parish, used during the Great Plague of 1665 to monitor the spread of the epidemic.
Biopsy	A diagnostic procedure developed in the twentieth century where a small sample of tissue or flesh is gathered from a patient to be examined by medical scientists in a laboratory, enabling highly accurate diagnosis.
Blood Depot	The precursor to modern blood banks, first established by the RAMC before the Battle of Cambrai in 1917, using sodium citrate and glucose to store type-O donor blood in advance of heavy casualties.

Term	Definition
Brodie Helmet	A steel combat helmet introduced by the British Army in 1915 to protect soldiers' heads from falling shrapnel and shell fragments, which dramatically reduced the rate of fatal head injuries.
Bronchoscopy	A high-tech diagnostic procedure where a thin, flexible tube with a camera is passed down a patient's airway into the lungs, allowing doctors to inspect the lung tissue and collect cell samples for biopsy.
Buboes	The painful, pus-filled swellings in the armpits or groin that were the defining physical symptom of bubonic plague.
Carrel-Dakin Method	An antiseptic wound treatment developed during WWI where a sterilized chemical solution was continuously pumped through narrow tubes placed deep inside open wounds to kill bacteria and prevent gas gangrene.
Casualty Clearing Station (CCS)	Well-equipped medical facilities located several miles behind the front line, close to transport links, where urgent surgery (such as amputations) and critical wound care took place.
Chain of Evacuation	The structured series of stages through which wounded soldiers were transported from the frontline trenches to increasingly secure and advanced medical facilities, prioritising speed to prevent infection.
Circulation	The movement of blood around the body through arteries and veins, pumped by the heart, which disproved Galen's ancient theory that blood was constantly produced by the liver.
Compulsory	Required by law or rule; a fundamental shift in government policy seen when the state made smallpox vaccination mandatory in 1852 and forced local councils to provide clean water and sewers in 1875.
Culture Medium	The nutrient-rich substance (such as agar jelly or liquid broth) used in laboratories to grow and cultivate microbes, which Alexander Fleming used to keep his penicillin samples alive.
Dissolution of the Monasteries	The action taken by Henry VIII in 1536 to close down Catholic religious institutions, which caused the immediate closure of the vast majority of hospitals in England since they were run by monks and nuns.
DNA	Deoxyribonucleic acid; the substance in human cells carrying genetic information that determines hereditary characteristics, discovered as a double helix in 1953 by James Watson and Francis Crick.
First Aid Nursing Yeomanry (FANY)	A voluntary women's organisation founded in 1907 whose members served on the Western Front as ambulance drivers, mobile kitchen staff, and nurses, providing vital logistical support to the RAMC.

Term	Definition
Four Humours	The ancient medical theory created by Hippocrates stating that the body was made of four vital liquids (blood, phlegm, yellow bile, and black bile) which had to be perfectly balanced to maintain good health.
Gas Gangrene	A rapid and deadly wound infection caused by anaerobic bacteria found in the heavily manured agricultural soil of Flanders, which thrived in deep, oxygen-depleted wounds caused by shrapnel.
Germ Theory	The breakthrough scientific discovery published by Louis Pasteur in 1861 proving that microscopic organisms (microbes) in the air cause decay and disease, which permanently disproved the theory of Spontaneous Generation.
Human Genome Project	A massive international scientific collaboration launched in 1990 and completed in 2001, which successfully mapped and decoded all three billion DNA pairs in the human body.
Humanism	A Renaissance intellectual movement characterized by a love of learning, a renewed interest in classical scholars, and the belief that human beings could use observation and reason to discover the truth for themselves rather than relying solely on religious authority or tradition.
Iatrochemistry	Also known as medical chemistry; a popular 17th-century practice inspired by Paracelsus that experimented with chemical preparations, metals, and minerals (like antimony or mercury) as cures rather than relying solely on traditional herbs.
Immunotherapy	A modern, advanced cancer treatment that works by boosting or stimulating the patient's own immune system, helping it to recognize, target, and destroy cancer cells.
Inoculation	The pre-19th century preventative practice of deliberately introducing pus from a mild smallpox scab into a cut on a healthy person's skin, which was expensive, risky, and could trigger a fatal outbreak of the disease.
Laissez-faire	The pre-19th century government attitude of minimal interference in the day-to-day lives of the population, based on the belief that public health and living conditions were not the responsibility of the state.
Magic Bullet	A chemical compound designed to target and kill specific disease-causing microbes inside the body without harming the patient's healthy cells, pioneered by Paul Ehrlich's Salvarsan 606 in 1909.
Miasma	The belief that bad, smelly air filled with harmful fumes from rotting organic matter or swamps could corrupt the body's humours and cause disease.

Term	Definition
Mobile X-ray Unit	A portable radiographic vehicle equipped with a dynamo generator to power an X-ray tube, used close to the frontline on the Western Front to locate shrapnel and bullets inside wounded soldiers.
Nanny State	A term used by critics to describe government policies and regulations that actively interfere with or restrict personal lifestyle choices (such as smoking or eating) in the name of protecting public health.
National Health Service (NHS)	A government-funded system launched in 1948 that made comprehensive medical treatment, hospital care, and primary consultations completely free at the point of access for all British citizens.
Nullius in Verba	The Latin motto of the Royal Society, meaning 'Take nobody's word for it', which encapsulated the Renaissance shift toward establishing scientific truth through experimentation and empirical evidence rather than traditional authority.
Phlebotomy	The medical term for bloodletting, which was the most common medieval treatment used to rebalance the humours by cutting a vein, using leeches, or cupping.
Quarantine	The practice of isolating the sick from the healthy to stop the spread of disease, such as locking infected families inside their homes for 40 days.
Regimen Sanitatis	A loose set of healthy living instructions written by physicians to help patients prevent illness through moderate diet, exercise, hygiene, and sleep.
Searchers	Local officials (usually older women) appointed by authorities during the Great Plague of 1665 to inspect corpses, identify plague deaths, and ensure infected households were quarantined.
Self-flagellation	The act of whipping oneself in public, practiced by groups of religious zealots (flagellants) to show God they were sorry for their sins in hopes of stopping the plague.
Shell Shock	The contemporary term (often recorded by medical staff as NYD.N, or 'Not Yet Diagnosed, Nervous') for the debilitating psychological trauma and mental breakdown caused by constant artillery bombardment.
Shrapnel	The fragments of metal casing scattered by exploding high-explosive artillery shells, which caused the vast majority of battlefield wounds and dragged dirty clothing and infected mud into the body.
Spontaneous Generation	The popular pre-1860s theory that microbes were the natural product, rather than the cause, of decay and decomposing matter.

Term	Definition
Spot Map	A detailed mapping technique created by John Snow during the 1854 Soho outbreak, where he drew marks on a street map to represent cholera deaths and visually trace the source of contamination to the Broad Street water pump.
Theory of Opposites	A medical treatment developed by Galen which aimed to restore balance to the body by applying the opposite quality to a dominant humour (e.g., treating a cold with hot peppers).
Therapeutic Agent	A chemical or biological substance developed to actively treat, cure, or heal a specific disease or infection inside a living patient, rather than just acting as a surface antiseptic.
Theriaca	A popular, complex spice-based herbal remedy containing up to 70 ingredients, widely used as a 'cure-all' in medieval England.
Thomas Splint	A metal leg splint designed before the war but introduced to the Western Front in 1916, which kept a broken leg rigid during evacuation, reducing the death rate from compound fractures from 80% to under 20%.
Transference	The Renaissance medical theory that an illness or fever could be physically transferred from a patient to another object or animal (e.g., rubbing warts with an onion or sleeping with a sheep).
Trench Fever	A highly infectious disease characterized by severe muscle pain and flu-like symptoms, spread among soldiers by body lice that nested in the warm seams of their clothing.
Trench Foot	A painful, disabling condition caused by standing for long periods in cold, wet, waterlogged mud, which caused feet to become numb, swollen, and prone to gangrene and amputation if not treated quickly.
Vaccination	The safer, scientific preventative technique pioneered by Edward Jenner in 1796 using mild cowpox matter to provide immunity against smallpox, named after 'vacca', the Latin word for cow.

2. Key Individuals Matrix

Hippocrates

Ancient Greek Physician

Actions: Created the Theory of the Four Humours (blood, phlegm, yellow bile, and black bile) and authored the Hippocratic Oath. Championed clinical observation, recording symptoms, and treating illnesses using natural therapies like the Theory of Opposites.

Impact: Shifted medicine away from supernatural or religious superstition toward a natural, rational explanation for disease. Established the foundational importance of patient observation and diagnostic note-taking.

Limitations: His theories were biologically incorrect. He lacked the technology to prove his ideas or observe microscopic pathogens. Once his writings were dogmatized by the Catholic Church, any attempt to question humoural theory was viewed as heresy, trapping medical progress in a static framework for over a thousand years.

Claudius Galen

Ancient Roman Physician

Actions: Combined Hippocratic humourology with his own animal dissections (on pigs and apes) to write over 350 books detailing human anatomy, physiology, and pathology. Developed the Theory of Opposites to balance the Four Humours.

Impact: Created a complete, rational, and natural system of medicine that replaced supernatural guesswork with a cohesive physiological model. His ideas were adopted as absolute dogma by the Catholic Church.

Limitations: Because he only dissected animals, he made major anatomical errors (e.g., claiming the human jaw is made of two bones, the liver has five lobes, and blood flows through invisible pores in the heart septum). His errors remained unchallenged for over 1,300 years because the Church banned human dissection and prosecuted free-thinkers.

Roger Bacon

13th-Century Scientist

Actions: An English scientist and Franciscan monk who argued that physicians should stop blindly memorizing classical texts (such as Galen's) and instead perform their own direct research, carry out experiments, and observe nature.

Impact: Promoted an early precursor to the empirical scientific method, challenging the scholastic, book-based complacency of medieval medical universities.

Limitations: His work was completely suppressed by institutional authority. In 1277, the Church imprisoned him for heresy and challenging traditional authority. Because book production was entirely controlled by monasteries, his experimental ideas were kept from circulating, resulting in zero short-term impact on daily medical care.

Andreas Vesalius

Pioneering Anatomist

Actions: Conducted direct, systematic human dissections of executed criminals and closely supervised standard, printed anatomical illustrations of his findings.

Impact: Published De Humani Corporis Fabrica (1543). Disproved over 200 of Galen's classical errors, proving that the human jaw is a single bone, the liver has two lobes, and the septum contains no pores. Turned anatomy into an empirical, observational science.

Limitations: His anatomical breakthroughs had no immediate clinical application; knowing the exact structure of the skeleton did not help physicians cure illnesses, meaning daily medical treatments remained unchanged. Many traditional physicians initially ignored or fiercely attacked his work.

William Harvey

Royal Physician

Actions: Dissected cold-blooded animals to observe slow heartbeats, and used a system of rods and ligatures on human arms to prove that blood flows only one way through veins.

Impact: Published De Motu Cordis (1628). Proved that the heart acts as a muscular pump circulating blood around the body, disproving Galen's theory that blood was constantly manufactured by the liver and absorbed by the tissues.

Limitations: His discovery had no direct clinical treatment value at the time, as knowing blood circulated could not cure disease. He lacked microscopes powerful enough to prove the existence of capillaries (which connect arteries and veins; discovered by Malpighi in 1661). Many contemporary doctors rejected his findings, and his ideas did not appear in English university textbooks until decades later.

Thomas Sydenham

The "English Hippocrates"

Actions: Rejected book-based dogmas and spent years closely observing patients at their bedsides, systematically recording and classifying their symptoms.

Impact: Published Observationes Medicae (1676). Proven to be the first to classify diseases as separate species, famously distinguishing scarlet fever from measles. Popularized natural treatments, such as prescribing fresh air, rest, and cinchona bark (quinine) for malaria.

Limitations: Although he revolutionized diagnosis, he was unable to explain the biological cause of disease because Germ Theory did not exist. He still believed in miasma and held traditional views, meaning many of his therapies remained observational rather than truly curative.

Edward Jenner

Pioneer of Vaccination

Actions: Observed that dairymaids who contracted cowpox were immune to smallpox. In 1796, he systematically tested this by inoculating a young boy, James Phipps, with cowpox, and later exposing him to smallpox.

Impact: Discovered the world's first vaccination. His discovery saved millions of lives and eventually led to the global eradication of smallpox.

Limitations: He had no scientific understanding of how or why vaccination worked because he had no knowledge of germs. Consequently, his breakthrough was a "one-off" that he could not scientifically adapt to prevent other deadly infectious diseases like cholera, typhoid, or tuberculosis.

Edwin Chadwick

Public Health Reformer

Actions: Researched urban slums and published the landmark 1842 Report on the Sanitary Conditions of the Labouring Population, proving that filthy living conditions caused disease, which trapped the working class in poverty.

Impact: His report served as the direct political catalyst for the passing of the 1848 Public Health Act, which established the first General Board of Health and proved that state-led environmental sanitation could prevent epidemics.

Limitations: He was a firm believer in the incorrect miasma theory (foul smells causing disease), meaning he focused on flushing sewage into rivers (contaminating drinking water). His autocratic style and "nanny state" bureaucracy generated fierce political and public opposition, leading the government to dissolve the Board of Health in 1854. Crucially, the 1848 Act was permissive (optional), meaning most cities refused to implement it.

John Snow

Surgeon and Epidemiologist

Actions: Mapped cholera deaths around the Broad Street pump in Soho during the 1854 outbreak, proving that fatalities were clustered around a single water source; removed the pump handle.

Impact: Proved that cholera was a water-borne disease, permanently disproving the dominant theory of miasma.

Limitations: Because Germ Theory had not yet been published, he could not provide the microscopic biological proof of what was actually in the water causing the cholera. Consequently, the government was extremely slow to act, taking over twenty years to pass the compulsory 1875 Public Health Act to clean up urban water supplies.

Florence Nightingale

Founder of Modern Nursing

Actions: Sanitized the military hospital at Scutari during the Crimean War (scrubbing wards, improving ventilation, washing sheets) and established the first professional training school for nurses.

Impact: Published Notes on Nursing (1859). Reduced Scutari's hospital mortality rate from 40% to 2%. Transformed nursing into a highly respected, professional career and popularized the hygienic "pavilion" layout for modern hospitals.

Limitations: She did not believe in Germ Theory, remaining a firm believer in the unscientific theory of miasma until her death. While her sanitary reforms successfully removed dirt and reduced infections, her unscientific beliefs meant she did not focus on antiseptic or aseptic surgical standards.

James Simpson

Pioneer of Anesthetics

Actions: Experimented with inhaling various chemical gases with his colleagues until they fell unconscious under his table from chloroform in 1847.

Impact: Discovered chloroform as a highly effective surgical anesthetic. Solved the barrier of physical pain in surgery, which was popularized after Queen Victoria used it during childbirth in 1853.

Limitations: His discovery triggered the "Black Period of Surgery" (1847–1870). Because patients were unconscious, surgeons attempted longer, deeper, and riskier internal operations; however, because they operated with dirty hands and tools, they carried deadly bacteria deep into wounds, causing infection rates to spike. Early patients also died from chloroform overdoses because safe dosages were hard to regulate.

Louis Pasteur

French Chemist

Actions: Conducted swan-neck flask experiments to prove that sterile liquids only decay when exposed to airborne microbes, rather than decay naturally generating them.

Impact: Published Germ Theory in 1861. Permanently shattered the unscientific theory of spontaneous generation. He later developed artificial vaccines for chicken cholera, anthrax, and rabies.

Limitations: He was a chemist, not a medical doctor, which led many members of the British medical establishment to reject his findings as irrelevant to human pathology. In his early work, he was unable to prove exactly which specific germ caused which specific human disease.

Joseph Lister

Pioneer of Antiseptic Surgery

Actions: Applied Pasteur's Germ Theory to wound healing, utilizing carbolic acid to spray the operating air, wash hands/instruments, and soak patient dressings.

Impact: Developed antiseptic surgery in 1865. Reduced post-operative mortality rates in his wards from 46% to 15%. His success paved the way for the development of entirely sterile, aseptic surgical environments by 1900.

Limitations: Carbolic acid was highly corrosive, irritating surgeons' hands and burning patients' lungs. Many contemporary doctors refused to use it because it made operations slower and more unpleasant, and because some did not yet believe in Germ Theory, they applied it incorrectly with poor results.

Robert Koch

Father of Bacteriology

Actions: Developed groundbreaking laboratory techniques: stained transparent microbes with industrial dyes, grew them in pure cultures on solid agar jelly, and photographed them.

Impact: Isolated the specific bacterial pathogens responsible for anthrax, tuberculosis (1882), and cholera (1883). Established "Koch's Postulates," a rigorous scientific checklist to prove microbe-disease links, transforming bacteriology into a clinical science.

Limitations: While he was brilliant at diagnosing the specific causes of disease, he could not cure them; his attempts to develop a cure for tuberculosis (tuberculin) failed, showing that identifying a pathogen did not immediately yield a chemical cure.

Wilhelm Roentgen

German Physicist

Actions: Discovered X-rays in 1895 by covering a cathode-ray tube with black cardboard, noticing that invisible rays could pass through solid objects and expose photographic plates to show bone structures.

Impact: Created the medical science of radiology. For the first time, doctors could diagnose internal fractures, bone diseases, and embedded foreign objects non-invasively, eliminating the need for exploratory surgery.

Limitations: Early X-ray technology was highly primitive and dangerous: exposure times were incredibly long (requiring patients to sit still for up to 90 minutes), the machines were massive and fragile (difficult to move), and they released toxic doses of radiation that caused severe skin burns and hair loss.

Marie Curie

Physicist and Chemist

Actions: Polish-French physicist who realized the urgent military need for radiology on the Western Front; equipped 20 mobile X-ray vans ("petites Curies") and trained 150 women to operate them.

Impact: Revolutionized battlefield triage by bringing diagnostic X-ray technology directly to forward Casualty Clearing Stations (CCS), allowing frontline surgeons to rapidly locate shrapnel and bullets before gas gangrene set in.

Limitations: Her mobile units were exceptionally fragile, required constant engine power from the vans to run, and struggled to navigate the destroyed, mud-choked terrain of the Ypres Salient or the Somme. Furthermore, a lack of protective shielding meant she and her operators suffered chronic, lethal exposure to radiation.

Karl Landsteiner

Biologist and Physician

Actions: Discovered blood groups (A, B, and O) in 1901 (with colleagues discovering AB in 1902), proving that donor and recipient blood must be matched to prevent fatal immune reactions.

Impact: Transformed blood transfusions from a highly lethal, outlawed gamble into a safe, scientifically controlled, and routine clinical procedure.

Limitations: He did not solve the critical problem of clotting and storage. Because blood clotted immediately outside the body, transfusions still had to be performed directly on a person-to-person basis with both donor and patient present, which was logistically impossible on a chaotic battlefield.

Paul Ehrlich

German Physician

Actions: Led a multidisciplinary research team that systematically tested hundreds of chemical compounds to find a synthetic dye that would target and destroy specific bacteria inside the body.

Impact: Developed Salvarsan 606 in 1909. This was the world's first synthetic chemical "magic bullet", which successfully cured syphilis.

Limitations: Salvarsan 606 was highly toxic because it was arsenic-based; it could easily kill the patient if administered in the wrong dosage. It was also highly specific, capable only of targeting the syphilis bacterium rather than treating a wide range of bacterial infections.

Sahachiro Hata

Japanese Bacteriologist

Actions: Japanese bacteriologist who joined Paul Ehrlich’s research team in Germany; systematically re-tested hundreds of rejected chemical compounds until finding that compound 606 successfully killed the syphilis spirochete.

Impact: Co-discovered Salvarsan 606 in 1909, the world's very first synthetic chemical "magic bullet" designed to target and destroy a specific microorganism inside the living human body.

Limitations: Salvarsan 606 was highly toxic because it was arsenic-based; if administered incorrectly, it could easily kill the patient rather than cure them. Additionally, it was extremely difficult to dissolve and inject, and it only cured syphilis, offering no protection against other common bacterial infections.

Gerhard Domagk

German Pathologist

Actions: Discovered Prontosil in 1932, a sulfonamide compound that successfully killed streptococcal bacteria inside the blood of living mice.

Impact: Developed the second magic bullet and the first commercially available sulfonamide drug, providing a highly effective cure for deadly systemic infections like puerperal fever, pneumonia, and scarlet fever.

Limitations: Prontosil was a bacteriostatic sulfonamide, meaning it stopped bacteria from multiplying but did not actively destroy them, relying instead on the patient's own immune system to clear the infection. It was entirely ineffective against deep, necrotic tissue wounds (which were common on the Western Front) and could cause severe kidney damage.

Alexander Fleming

Scottish Bacteriologist

Actions: Left a petri dish of staphylococcus bacteria open on his laboratory bench before going on holiday in 1928, returning to find a mould (Penicillium notatum) had contaminated the dish and destroyed the surrounding bacteria.

Impact: Discovered the antibacterial properties of penicillin and published his findings in 1929. His paper served as the scientific starting point for subsequent antibiotic research.

Limitations: He was unable to extract, purify, or stabilize the active penicillin compound. Believing it was too unstable to be used as anything more than a minor local antiseptic, he abandoned his research in 1931, leaving it completely undeveloped as a systemic drug.

Howard Florey & Ernst Chain

Pharmacologist & Biochemist

Actions: Revived Fleming's research at Oxford University in 1938–1939; assembled a specialist team of biochemists and pathologists to purify, stabilize, and test penicillin on mice and humans.

Impact: Successfully developed penicillin into the world's first stable, usable, and systemic antibiotic by 1940–1941.

Limitations: They lacked the massive funding, industrial equipment, and factory space needed to mass-produce the drug in wartime Britain. Early patients (such as Albert Alexander in 1941) died when their supply of penicillin ran out, proving that scientific success was useless without industrial mass production.

James Watson & Francis Crick

Molecular Biologists

Actions: Published the first accurate molecular model of the double-helix structure of DNA in 1953, utilizing X-ray diffraction data and physical cardboard models to solve the genetic puzzle.

Impact: Unlocked the genetic paradigm of modern medicine, explaining how hereditary characteristics are passed down and directly enabling the launch of the Human Genome Project (1990–2003).

Limitations: Their discovery was purely theoretical. Understanding the physical structure of DNA did not translate into immediate cures; it took decades of research to link genes to specific diseases like lung cancer, and medical science still lacks the technology to safely and universally edit genetic mutations today.

Rosalind Franklin

X-ray Crystallographer

Actions: Captured Photo 51 (1952), a highly precise X-ray crystallography photograph of crystallized DNA fibers, and calculated the physical dimensions of the helix.

Impact: Provided the essential, mathematical proof of DNA's double-helix structure, which Watson and Crick relied upon to build their successful molecular model.

Limitations: Her impact was severely limited by systemic sexism in the 1950s academic community; her data was shared without her permission, and she was denied co-authorship on the initial discovery paper. Because her expertise was strictly in chemical physics, she played no role in translating the discovery of DNA into clinical medical therapies.

Hugh Owen Thomas

Orthopaedic Surgeon

Actions: Designed the Thomas splint—a rigid metal leg frame featuring a leather ring at the hip to pull the leg under traction and keep the femur fully immobilized.

Impact: When adopted on the Western Front in 1916, his splint revolutionized the treatment of femur fractures, dropping the mortality rate from compound leg wounds from 80% to under 20% by stopping internal bleeding, bone movement, and shock.

Limitations: He designed and used the splint in his civilian practice decades before the First World War, but his work was entirely ignored by the British military and medical establishment during his lifetime due to professional conservatism. It was only the desperate, high-mortality crisis of trench warfare that forced its retrospective, widespread adoption in 1916.

Harold Gillies

Pioneer of Plastic Surgery

Actions: Established a specialized reconstructive surgery unit at the Queen's Hospital in Sidcup, Kent, to treat British soldiers who had suffered horrific facial disfigurements from shrapnel and machine-gun fire.

Impact: Pioneered advanced plastic surgery techniques, notably inventing the tube pedicle (which preserved blood flow to skin grafts) and advanced bone grafting.

Limitations: Reconstructive surgery was incredibly slow and painstaking, requiring dozens of separate, highly painful operations over several years. Because antibiotics did not exist yet, patients faced an extremely high risk of contracting fatal infections during these prolonged procedures.

Oswald Hope Robertson

Medical Researcher

Actions: Set up a forward supply depot of preserved donor blood before the Battle of Cambrai in 1917.

Impact: Created the world's first functional blood bank. Utilized sodium citrate to prevent clotting and dextrose/glucose to preserve red blood cells, allowing universal Group O blood to be safely stockpiled for up to 21 days.

Limitations: Preserved blood could only be kept fresh for a maximum of three weeks and required delicate, constant refrigeration in basic wooden crates packed with ice, which was exceptionally difficult to maintain on a chaotic, mud-soaked battlefield.

Sir Robert Jones

Orthopaedic Surgeon

A pioneering British orthopaedic surgeon and nephew of Hugh Owen Thomas. During WWI, he recognised the life-saving potential of his uncle's invention and drove the mass deployment of the Thomas Splint to the Western Front in late 1915. His insistence that frontline stretcher-bearers be trained to apply the splint before moving patients successfully reduced the mortality rate of compound femur fractures from 80% to 20%.

Dr. Harvey Cushing

American Neurosurgeon

An innovative American neurosurgeon who served with the British Expeditionary Force and later the American forces on the Western Front. He pioneered new techniques for brain surgery under battlefield conditions, notably experimenting with using large magnets to extract deep-seated steel shrapnel fragments from the brains of wounded soldiers, and meticulous documentation of head wounds.

Pat Beauchamp

FANY Ambulance Driver

A prominent volunteer in the First Aid Nursing Yeomanry (FANY). She served as an ambulance driver on the Western Front, proving the crucial value of motorized transport in rapid casualty evacuation. Despite losing her leg in a severe accident in France, she continued to serve and later wrote the memoir 'Fanny Goes to War', highlighting the physical dangers and vital contributions of female volunteers.

3. Exam Question Bank

- [KT1.1: What did Medieval people believe caused illness?] Explain why there was so little progress in understanding the causes of disease during the Middle Ages. (12 marks)
- [KT1.1: What did Medieval people believe caused illness?] Q2. ‘The Theory of the Four Humours was the main idea about the cause of disease in the Middle Ages’. How far do you agree? Explain your answer. (16 marks) You may use the following in your answer:
  • Galen
  • the Church
  You must also use information of your own.
- [KT1.2: How did Medieval people try to prevent and treat disease?] Explain why bleeding and purging were such common treatments in the medieval period (c1250–c1500). (12 marks)
- [KT1.2: How did Medieval people try to prevent and treat disease?] ‘In the years c1250–c1500, the physician was the most important person providing care and treatment.’ How far do you agree? Explain your answer. (16 marks + 4 marks for SPaG)
- [KT1.3: How did people respond to the Black Death?] Explain why the Black Death spread so rapidly in Britain in 1348. (12 marks)
- [KT1.3: How did people respond to the Black Death?] Q2. ‘The main reason why medical care and treatment was ineffective during the medieval period was because medical knowledge was based on Galen's ideas.’ How far do you agree? Explain your answer. (16 marks) You may use the following in your answer:
  • bloodletting

• dissection
You must also use information of your own.

- [KT2.1: Did the Renaissance change beliefs about the causes of illness?] Explain why there was rapid progress in anatomical knowledge in the Renaissance period (c1500–c1700). (12 marks)
- [KT2.1: Did the Renaissance change beliefs about the causes of illness?] Explain one way in which knowledge of human anatomy in the Renaissance period (c1500–c1700) was different from knowledge of human anatomy in the medieval period (c1250–c1500). (4 marks)
- [KT2.1: Did the Renaissance change beliefs about the causes of illness?] Explain why there were changes in the way ideas about the causes of disease and illness were communicated in the period c1500–c1700. (12 marks)
- [KT2.2: Did treatments improve during the Renaissance?] Explain why there was little change in medical treatments during the Renaissance period (c1500–c1700), despite new anatomical discoveries. (12 marks)
- [KT2.2: Did treatments improve during the Renaissance?] Explain one way in which medical treatments in the Renaissance period (c1500–c1700) were similar to medical treatments in the medieval period (c1250–c1500). (4 marks)
- [KT2.2: Did treatments improve during the Renaissance?] Explain why there was rapid progress in anatomical knowledge in the Renaissance period (c1500–c1700). (12 marks)
- [KT2.3: How significant were William Harvey and the Great Plague?] Explain why the government's response to the Great Plague of 1665 was more effective than its response to the Black Death of 1348. (12 marks)
- [KT2.3: How significant were William Harvey and the Great Plague?] Explain one way in which the response to the Great Plague in London (1665) was different from the response to the Black Death (1348). (4 marks)
- [KT2.3: How significant were William Harvey and the Great Plague?] ‘The most effective method of preventing the spread of the Great Plague (1665) was the use of quarantine.’ How far do you agree? Explain your answer. (16 marks + 4 marks for SPaG)
- [KT3.1: What breakthrough discoveries changed our understanding of disease causes?] Explain why Louis Pasteur’s Germ Theory (1861) was a major turning point in medicine. (12 marks)
- [KT3.1: What breakthrough discoveries changed our understanding of disease causes?] Explain one way in which ideas about the cause of illness in the period c1700–c1900 were different from ideas about the cause of illness in the period c1500–c1700. (4 marks)
- [KT3.1: What breakthrough discoveries changed our understanding of disease causes?] Q2. ‘Louis Pasteur's publication of the Germ Theory was the biggest turning point in understanding the causes of disease.’ How far do you agree? Explain your answer. (16 marks) You may use the following in your answer:
  • spontaneous generation
  • Robert Koch
  You must also use information of your own.
- [KT3.2: How did the Industrial Revolution transform prevention and treatment?] Explain why Joseph Lister’s development of carbolic acid was so significant in the history of surgery. (12 marks)

- [KT3.2: How did the Industrial Revolution transform prevention and treatment?] Explain one way in which surgical treatments in the period c1700–c1900 were different from surgical treatments in the medieval period (c1250–c1500). (4 marks)
- [KT3.2: How did the Industrial Revolution transform prevention and treatment?] 'The discovery of Carbolic Acid was the most significant turning point in surgery in the years c1700–c1900.' How far do you agree? Explain your answer. (16 marks + 4 marks for SPaG)
- [KT3.3: How significant were Edward Jenner and John Snow?] Explain why there was rapid progress in public health during the second half of the 19th century. (12 marks)
- [KT3.3: How significant were Edward Jenner and John Snow?] Explain one way in which the role of the government in preventing disease in the period c1700–c1900 was different from the period c1500–c1700. (4 marks)
- [KT3.3: How significant were Edward Jenner and John Snow?] Explain why there was rapid progress in public health in the second half of the nineteenth century (c1850–c1900). (12 marks)
- [KT4.1: How have modern discoveries changed our understanding of illness?] Explain why the discovery of the structure of DNA was a significant breakthrough in understanding the causes of illness. (12 marks)
- [KT4.1: How have modern discoveries changed our understanding of illness?] Explain one way in which ideas about the cause of illness in the modern period (c1900–present) were different from ideas in the period c1700–c1900. (4 marks)
- [KT4.1: How have modern discoveries changed our understanding of illness?] Q2. 'The discovery of DNA was the most significant breakthrough in understanding the causes of illness in the period c1900–present.' How far do you agree? Explain your answer. (16 marks) You may use the following in your answer:
  • Watson and Crick
  • Lifestyle factors
  You must also use information of your own.
- [KT4.2: How have prevention and treatment advanced in the modern era?] Explain why there was rapid progress in the development of penicillin in the 1940s. (12 marks)
- [KT4.2: How have prevention and treatment advanced in the modern era?] Explain one way in which the treatment of illness in the modern period (c1900–present) was different from the treatment of illness in the period c1700–c1900. (4 marks)
- [KT4.2: How have prevention and treatment advanced in the modern era?] 'Alexander Fleming was the most important individual in the development of penicillin.' How far do you agree? Explain your answer. (16 marks + 4 marks for SPaG)
- [KT4.3: How was Penicillin discovered and mass-produced?] Explain why the government took a more active role in preventing disease in the period c1900–present. (12 marks)
- [KT4.3: How was Penicillin discovered and mass-produced?] Explain one way in which ideas about the prevention of illness in the modern period (c1900–present) were different from the medieval period (c1250–c1500). (4 marks)
- [KT4.3: How was Penicillin discovered and mass-produced?] Explain why there was rapid progress in disease prevention in the period c1900–present. (12 marks)

- [KT4.4: How has the government tackled the modern epidemic of Lung Cancer?] Explain why the creation of the National Health Service (NHS) in 1948 improved public health in Britain. (12 marks)
- [KT4.4: How has the government tackled the modern epidemic of Lung Cancer?] Explain one way in which hospital care in the modern period (c1900–present) was different from hospital care in the period c1700–c1900. (4 marks)
- [KT4.4: How has the government tackled the modern epidemic of Lung Cancer?] Explain why the government took action to improve public health in the period c1900–present. (12 marks)
- [KT5.1: What was the historical and medical context of the Western Front?] Describe one feature of the First Battle of Ypres. (2 marks)
- [KT5.1: What was the historical and medical context of the Western Front?] Describe one feature of the Battle of the Somme. (2 marks)
- [KT5.1: What was the historical and medical context of the Western Front?] 1 (a). How useful are Sources A and B for an enquiry into the difficulties faced by soldiers in the trench system of the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)
- [KT5.2: How did the trench environment create new medical challenges?] Describe one feature of the trench system. (2 marks)
- [KT5.2: How did the trench environment create new medical challenges?] Describe one feature of trench foot. (2 marks)
- [KT5.2: How did the trench environment create new medical challenges?] 2 (a). How useful are Sources A and B for an enquiry into the impact of the terrain on the transport of the wounded on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)
- [KT5.3: What new illnesses and wounds did soldiers face?] Describe one feature of the effects of poison gas. (2 marks)
- [KT5.3: What new illnesses and wounds did soldiers face?] Describe one feature of shrapnel wounds. (2 marks)
- [KT5.3: What new illnesses and wounds did soldiers face?] 2 (a). How useful are Sources A and B for an enquiry into the effects of poison gas attacks on soldiers on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)
- [KT5.4: How did the RAMC and FANY operate the chain of evacuation?] Describe one feature of the FANY. (2 marks)
- [KT5.4: How did the RAMC and FANY operate the chain of evacuation?] Describe one feature of the Chain of Evacuation. (2 marks)
- [KT5.4: How did the RAMC and FANY operate the chain of evacuation?] 2 (a). How useful are Sources A and B for an enquiry into the problems of transporting the wounded on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)
- [KT5.5: What incredible medical advances were forged on the Western Front?] 2 (a). How useful are Sources A and B for an enquiry into the effectiveness of new medical developments used to treat wounded soldiers on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)